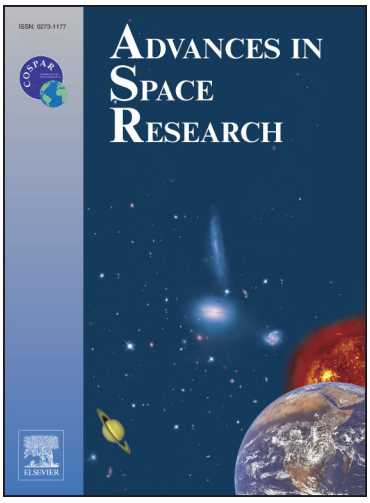
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A novel automated technique based on ensemble learning for prediction of soil moisture using satellite images

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Abstract

It is essential to comprehend how the environment and climate vary. The geophysical data of NASA under the mission of soil moisture active passive (SMAP) have shown to be a valuable resource, offering hitherto undiscovered insights into the dynamics of soil moisture, and associated geophysical terms. To evaluate the effect of global warming on climatic changes, analysis of the SMAP has been conducted in several regions of Gujarat, India, over the last four years (2020-2023) in the current study. After preprocessing the original dataset for missing value imputation, significant features with a greater influence on the surface soil moisture were extracted using a Random Forest (RF). A model for predicting soil moisture was constructed utilizing the selected features and the gradient boosting (GB) technique. The Gradient Boosting model achieved outstanding predictive accuracy (R^2 = 0.99 with minimal error), consistently surpassing the performance of other methods, including Lasso Regression, Linear Regression, and Random Forest. Though RF regressor was higher computational resource consuming than GB. The suggested predictive model has a lot of practical applications, including drought and climate monitoring.

Keywords: SMAP; Soil moisture; Regression; Gradient Boosting

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1. Introduction

Water is a very necessary element of the planet, playing a central role in sustaining ecosystems and human development. However, increasing urbanization and desertification continue to pose serious challenges to its availability and management (Abahussain et al., 2002). With the global population expected to reach 9.725 billion by 2050 (Wang & Moriarty, 2018), the demand for both food and water is intensifying. Efficient water resource management is now more critical than ever. Soil moisture plays a key role in this context. It acts as a temporary water reserve that directly supports both agriculture and climate regulation. Accurate knowledge of soil moisture levels is essential for tasks like irrigation scheduling (Hatfield et al., 2001; Gu et al., 2020; Gu et al., 2021), but conventional methods such as gravimetric sampling, tensiometers, neutron probes, and

reflectometry tools (Romano, 2014; Dobriyal et al., 2012; Babaeian et al., 2019) are often invasive, time-consuming, providing partial spatial coverage. To overcome these limitations, remote sensing and satellite-based techniques have emerged as transformative solutions over the past few decades (Singh & Gaurav, 2023; Müllerová et al., 2021). Specifically, surface soil moisture (0-5 cm depth) and root zone moisture (0-200 cm depth) are now measurable at large scales using satellite missions such as NASA's Soil Moisture Active Passive (SMAP) (SMAP, 2015; McColl et al., 2017). SMAP employs both microwave radiometer and synthetic aperture radar (SAR) technologies. These instruments enable the mission to capture global-scale soil moisture variations, including areas where ground access is challenging (Renju & Raju, 2023; Yueh, 2000). Recent developments in artificial intelligence have further advanced soil moisture modeling (Singh et al., 2025). Various well-known learning models like CNNs and ConvLSTM have been used to analyze satellite images and understand complex moisture-related patterns in soil (Xu et al., 2021; Liu et al., 2022). The earlier models are often computationally intensive when applying to environmental policy or agricultural planning. To address these challenges, the study provides an option which is more efficient and interpretable. An ensemble learning framework that combines Gradient Boosting (GB) Regressor and Random Forest (RF) has been utilized for surface soil moisture prediction. Rather than relying on image data, this model uses structured tabular data derived from SMAP's geophysical products. Covering the period from 2020 to 2023 and focusing on January, March, May, and August months that represent seasonal variation, this dataset captures soil-climate interactions across winter, summer, and monsoon periods. The selected study sites in Gujarat, India, include regions with diverse climatic and geographic characteristics such as semi-arid zones, coastal plains, industrial hubs, and arid landscapes. This diversity in the dataset enhances model generalization and provides a robust testing ground for the proposed method. By focusing on computationally lightweight, feature-driven modeling, this work aims to bridge the gap between scientific accuracy and real-world usability in soil moisture prediction (Lakra et al., 2024). A notable strength of this study is the way it integrates RF for identifying key features with GBR to carry out the final prediction of soil moisture. This two-step strategy allows for both interpretability and precision. The training of model is done on the tabular data of NASA SMAP mission, covering four different months across multiple seasons from 2020 to 2023. Also, data is collected from seven locations in Gujarat including Bhuj, Idar, Ankleshwar, Bhavnagar, Jamnagar, Vapi, and Palanpur, each offering a unique environmental and climatic profile. This level of spatial and temporal variability makes the proposed framework flexible and adaptable across a wide range of conditions.

2. Study locations

The study location covers a total of seven unique regions within the Gujarat state of India. These places were chosen because they offer a wide variety of environmental and meteorological circumstances, and as a result, they can give useful perceptions into the dynamics of moisture content in soil and management strategies. **Fig.1** shows the exact locations of study area in Gujrat, India. The proposed model was tested using the data from all locations shown in **Fig.1**. The

detailed description of the study locations is as follows (*BISAG-N*, 2023). Despite these improvements, a few points still warrant clarification regarding the specific characteristics of each region. A detailed description of each of the regions is as follows.

2.1. Bhuj frequently struggles with a lack of available water because of the city's location in a semi-arid climate. Due to the unpredictability of the weather patterns, Bhuj is an ideal place for the testing of innovative strategies for the management of soil moisture. Recent reports show that only half of the region's water bodies are still functioning, which highlights the need for good water resource management. Because Bhuj goes through significant climatic fluctuations, going from dry to rainy circumstances, it is an excellent location for the investigation of the dynamics of soil moisture.

2.2. Idar is in the middle of Gujarat, features a one-of-a-kind biological landscape that is distinguished by its hilly topography. This area has shifted in soil moisture that is caused by both topographical and climate factors acting in concert. The selection of Idar as one of the study locations adds to a better knowledge of how different factors of the landscape influence the amount of moisture in the soil.

2.3. Ankleshwar is an industrial hub in the state of Gujarat, Ankleshwar faces unique issues regarding the management of soil moisture. Ankleshwar is an attractive study location because of the interaction between industrial operations and environmental elements. The present study will provide light on the complicated dynamics of soil moisture in urban and industrial contexts.

2.4. Bhavnagar which is situated in close proximity to the ocean and contends with the consequences of its saline soil. Insights into the interrelationship between soil moisture dynamics and coastal variables are invaluable due to the exceptional conditions that exist in this region.

2.5 Jamnagar is widely recognized for its profusion of plant and animal life and showcases an ecosystem where soil moisture levels are intricately intertwined with ecological patterns and flora. It has a good interconnection among environmental factors, vegetation, and soil moisture.

2.6. Vapi has an environment, which is characterized by fluctuating levels of soil moisture, which is exacerbated by the vicinity of significant industrial sectors and bodies of water. By examining the intricate interplay between diverse industrial processes and the influence of water supplies on soil moisture, this research site proves to be valuable.

2.7. Palanpur is located in Northern Gujarat. It is distinguished by its deserting conditions and climate that is notably dissimilar. In arid regions, where water resources are of the utmost importance, this research site aids comprehension of soil moisture management.

Collectively, these seven regions offer a varied and all-encompassing framework for the present study, covering a wide range of environmental conditions and challenges in their entirety. The understanding of moisture content in soil dynamics and its inferences for the controlling of the environment in a natural manner can be significantly advanced by the insights obtained from selected study locations. A summary of some valuable information about the selected study locations is provided in **Table 1**.

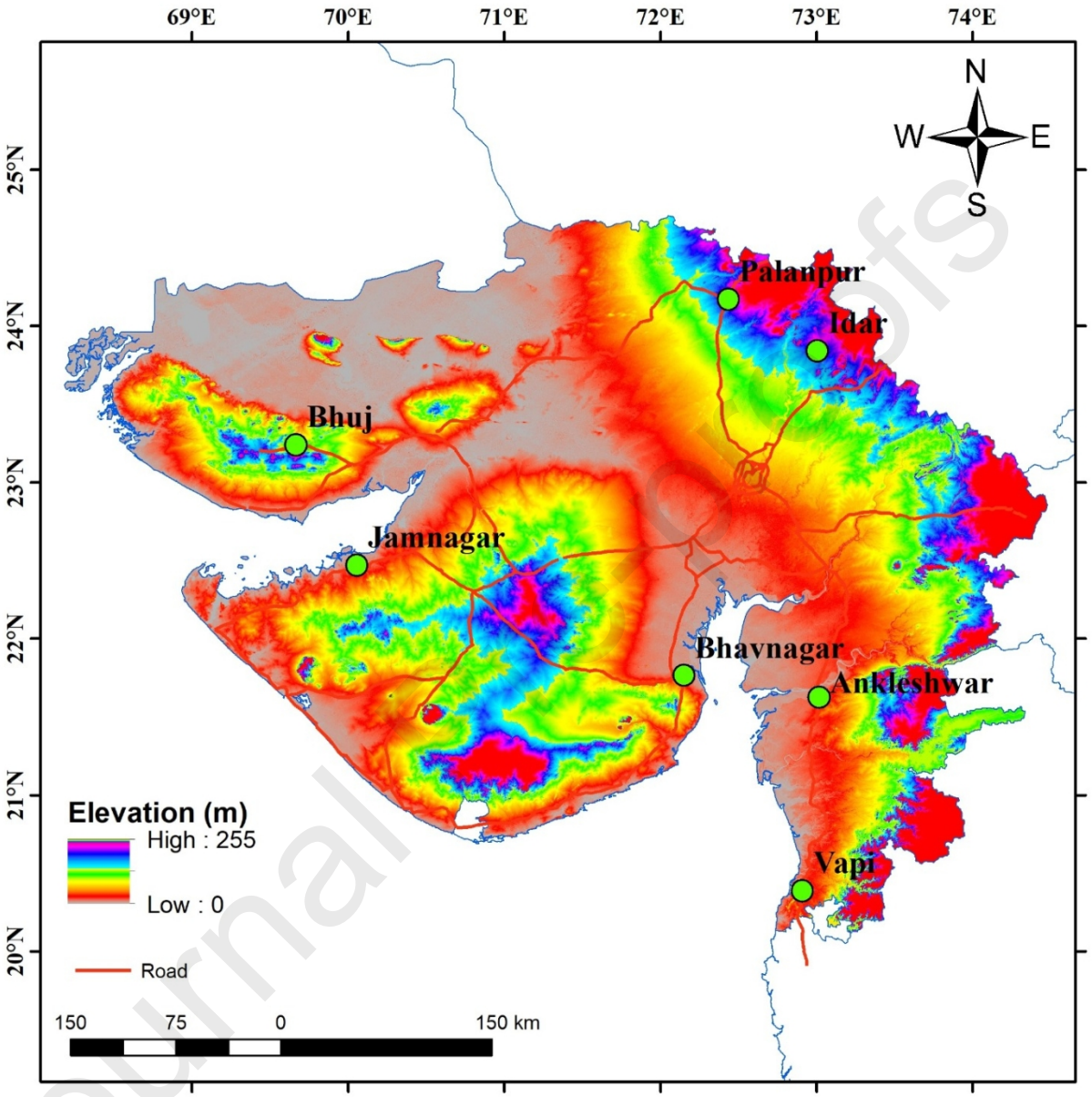


Fig. 1. Locations of study area.

Table 1 A brief info about the selected regions (*BISAG-N*, 2023).

Study location	Latitude	Longitude	Geographic Features	Climate Characteristics	Environmental Significance

Bhuj	23.2531	69.6693	Semi-arid	Borderline hot desert climate	Water scarcity
Idar	23.8481	73.0024	Hilly terrain	Tropical savanna climate	Landscape impact
Ankleshwar	21.6019	73.0032	Industrial hub	Hot semi-arid climate	Drought may common
Bhavnagar	21.7645	72.1519	Coastal region	Saline soil	Coastal soil dynamics
Jamnagar	22.4707	70.0577	Biodiversity	Ecosystem relationships	Vegetation dynamics
Vapi	20.3893	72.9112	Industrial region	Water proximity	Industrial impact
Palanpur	24.1721	72.4286	Arid conditions	Arid climate	Water resource significance

3. Methodology

3.1 SMAP Dataset

An open-source data of the satellite mission by NASA, SMAP (SMAP, 2015) clearly shown in **Table A2** (**Appendix A**) is selected for the present study. Specially, SMAP-L4 global 3-hourly 9 KM EASE-grid surface and root zone soil moisture geophysical data V007 has been selected. SMAP is a NASA initiative aimed at offering observations of soil moisture, which plays a significant role in study for soil moisture monitoring and its prediction. This dataset employs active radiometer as well as passive microwave measurements to estimate soil moisture levels on the earth's surface. The SMAP L4_SM data is prepared using satellite observations by SMAP mission as well as land surface observation by Goddard Earth Observing System Model, Version 5 (GEOS-5). Consequently, while surface soil moisture is directly observed by the SMAP radiometer, other geophysical variables such as rootzone moisture, soil temperatures at different depths, and runoff fluxes are derived through model simulations. This hybrid approach ensures a comprehensive and physically consistent dataset suitable for large-scale soil moisture analysis (Reichle, 2014; Rouzegari et al., 2025). The SMAP has been validated through multiple independent studies, which supports its reliability for large-scale soil moisture analysis (Lu et al., 2023; Albano et al., 2024; Wang et al., 2025). The SMAP-L4 global dataset includes geophysical, spatial, and temporal

information essential for soil moisture prediction. The geophysical variables include soil moisture, land surface temperature, and vegetation parameters, while spatial features such as coordinate grids are represented through latitude, longitude, and projection data. A full list of the dataset fields and their descriptions is provided in **Table A2 (Appendix A)**. Despite this, because geographical data contains all the features that affect soil moisture, it is employed in the subsequent analysis. SMAP uses a satellite that gathers eight datasets of a single place every three hours. Each dataset consists of 45 columns (each column representing features affecting soil moisture and value of soil moisture) and 6262144 rows (each row representing unique image information). The soil moisture surface (sm_surface) has 4577419 null rows (SMAP, 2015). Consequently, the analysis has only made use of 1684725 rows and 45 columns of datasets of January, March, May, and August months of 2020, 2021, 2022 and 2023 for seven locations. Additionally, one dataset is chosen at random each month, for a total of 16 datasets collected from a single site. In the current study, a total of 112 datasets has been collected and examined across 7 locations.

3.2 Data Preprocessing

The first stage in using SMAP is data preparation. Managing and inputting the missing data is one of the primary responsibilities of the preprocessing step (Hegazi et al., 2021). If missing values are not handled appropriately, a bias could be introduced into the knowledge (Wang and Singh, 2021; Chen et al., 2020). Instances of null or absent values frequently occur in geophysical data sets. It was crucial to address this challenge to ensure the quality of input data (Emmanuel et al., 2021). To systematically resolve this issue, this study has developed a Python script that utilizes the h5py module to use HDF5 files, which are commonly employed for data storage. The algorithm implemented in this study is designed to analyze SMAP files, which contain critical geophysical information. The subsequent items delineate the primary responsibilities executed by the script.

- The datasets are efficiently preprocessed by the script through the repeated management of multiple dataset files via a list of file paths, where each path represents a distinct dataset file.
- The main goal was to particularly deal with null values in the 'sm_surface' variable, which is important for soil moisture prediction. Null values in geophysical datasets are commonly represented as -9999.0. To solve this issue, script does the following.
  - A boolean mask named 'valid_rows_mask' is generated, which identifies rows where 'sm_surface' values do not equal -9999.0, effectively marking rows with valid data. Consequently, rows with missing 'sm_surface' values are systematically removed from the dataset.
  - The script offers valuable feedback throughout processing, displaying the structure of the dataset, the mask of valid rows, and alerts indicating the removal of null rows from each processed file.
- Further in rest of the features other than the target ‘sm_surface’, mean imputation is used to impute the missing value by performing column-wise mean imputation using Eq. 1.

$$Imputed_Value = 1/n \sum_{i=1}^{n} x_i \quad (1)$$

where ‘n’ is total number of values in the feature, x_i is the value of the feature.
After preprocessing, the 44 features are shown in **Table A1** (**Appendix A**).

3.3 Embedded Approach of Feature Selection

Data quality affects the performance of any machine learning model. Even the advance model cannot perform well if the data is noisy (Hegazi et al., 2021). A SMAP-L4 dataset initially contained 44 environmental variables (**Appendix Table A1**). To enhance soil moisture predictive capacity, an embedded feature selection method based on RF regression was employed. An exhaustive investigation into feature significance was conducted using the RF regressor, a robust ensemble learning method that combines multiple decision trees to improve predictive accuracy. Each decision tree was trained using the bootstrap aggregating (bagging) method, enhancing generalization (Khaire & Dhanalakshmi, 2022). '*L*' leaves on a decision tree split the feature space into '*L*' regions, denoted as R_l for each region, where *1 ≤l≤ L*. For a single tree, the function *f(p)* for prediction is given as (Chen et al., 2020)

$$f(p) = \sum_{l}^{L} c_l \mu(x, R_l) \qquad (2)$$

where 'L' is the number of regions in the feature space, 'R_l' is a region appropriate to 'l' and 'c_l' is a constant suitable to l.

$$\mu(x, R_l) = \begin{cases} 1 & if\ x \in R_l \\ 0 & otherwise \end{cases} \qquad (3)$$

where $\mu(x, R_l)$ in Eq. 3 is an indicator function which tells that the specific feature falls in the precise tree of the RF or not. The majority vote determines the final classification result. The objective of the embedded feature selection was to systematically identify the most influential environmental parameters affecting surface soil moisture prediction. Initially, all the 44 features present in the geophysical dataset were used to train the RF Regressor model. The importance score of features was computed based on the Ginni importance criterion, which evaluates the contribution of each feature towards reducing prediction error. RF is particularly effective in handling high-dimensional datasets (Khaire & Dhanalakshmi, 2022). Thus, the most critical features were selected to optimize model accuracy and computational efficiency (Chen et al., 2020). This approach effectively handles complex feature interactions and provides reliable importance ranking (Motta et al., 2021). Redundant and non-informative features were eliminated, retaining only significant predictors (Li et al., 2012). The RF model used for feature selection employed default hyperparameters from the scikit-learn library, including 100 estimators (n_estimators = 100), no restriction on tree depth (max_depth = None), minimum two samples required for a split (min_samples_split = 2), and square root feature sampling (max_features = 'sqrt'). Random state was not set during feature selection to allow natural randomness in tree formation. A schematic diagram of embedded feature selection method (Liu et al., 2019) is shown in **Fig. 2**. Based on the calculated importance scores, the top 14 features were selected.

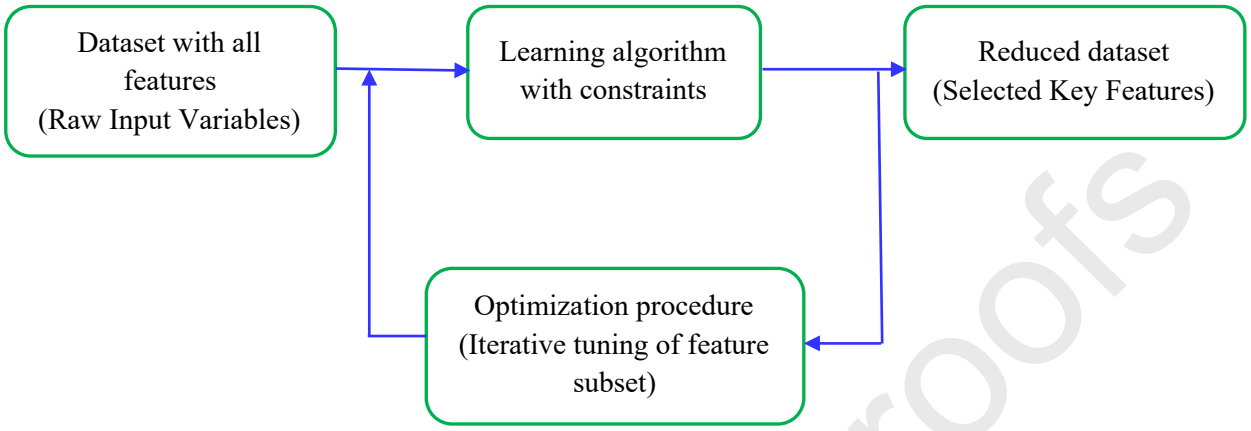


Fig. 2. Feature selection method.

The number 14 was finalized based on the importance score of features: selecting fewer than 14 features resulted in a noticeable decline in model performance, while selecting beyond 21 features did not yield significant gains as clearly shown in **Fig. 3** which shows the low error mean squared error (MSE) in model prediction when feature count is in between 14 to 21 and greater otherwise. The workflow of this feature selection process is shown in **Fig. 2** and the correlation between the selected features, and the surface soil moisture target variable is shown in **Fig. 4**. It is important to note that although several input variables are related to moisture, they represent distinct physical aspects of the land surface. For example, sm_rootzone looks at how much moisture is present deeper in the soil (around 0–100 cm), while sm_surface only covers the top layer (0–5 cm). The variables sm_surface_wetness and sm_rootzone_wetness don't give exact moisture levels but rather indicate how wet those layers generally are. While they might sound similar, RF and correlation plots were checked and found that they weren't overlapping. Each one added something different to the model. The most significant 14 features are shown in **Table 2.**

Table 2 List of 14 Features selected for soil moisture prediction using Random Forest.

Feature Name	**Description**
sm_surface_wetness	Wetness condition at surface (0–5 cm)
land_fraction_unsaturated	Fraction of land area unsaturated by water

sm_rootzone	Soil moisture in rootzone (0–100 cm)
overland_runoff_flux	Rate of surface runoff over land
precipitation_total_surface_flux	Total precipitation reaching land surface
windspeed_lowatmmodlay	Near-surface wind speed (model surface layer)
heat_flux_ground	Heat fluxes between ground and atmosphere
vegetation_greenness_fraction	Fraction of green vegetation cover
soil_temp_layer3	Soil temperature at 15–35 cm depth
soil_temp_layer5	Soil temperature at 60–100 cm depth
net_downward_longwave_flux	Net incoming longwave radiation at land surface
sm_rootzone_wetness	Wetness condition in rootzone soil layer
depth_to_water_table_from_surface_in_peat	Depth from surface to water table in peatlands
land_fraction_saturated	Fraction of land area saturated with water

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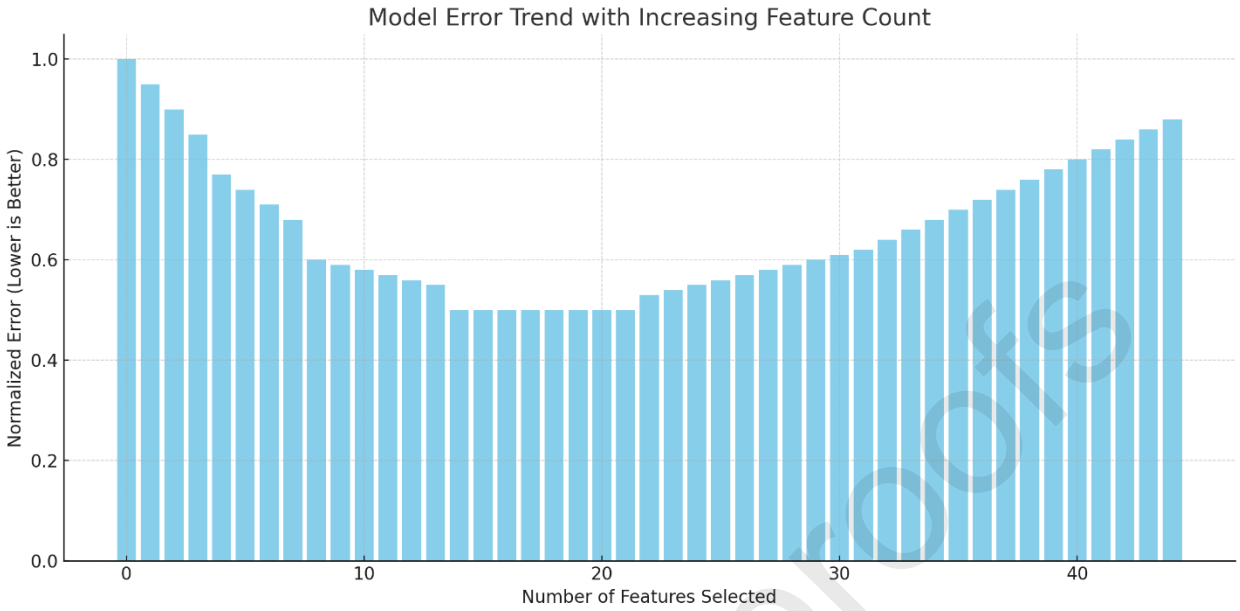


Fig. 3. Variation in model error (MSE) with number of selected features.

3.4 Gradient Boosting Regression for soil moisture prediction

The Gradient Boosting (GB) Regressor was employed to predict the surface soil moisture by using the 14 important features selected through the RF embedded feature selection method. GBR builds an ensemble model by sequentially adding decision trees, where each new tree minimizes the residual errors made by the previous ensemble (Rao et al., 2019). It is a combined model with enhanced stability and greater performance (Shrivastav & Jha, 2021). It uses the boosting algorithm to solve the regression problem. GBR minimizes a loss function using gradient descent (Li et al., 2018). The prediction after adding M trees is mathematically represented as

$$f_M(x_j) = \sum_m^M y_m h_m(x_j) \quad (4).$$

where $h_m(x_j)$ represents the prediction from the mth weak learner and y_m is its weight.

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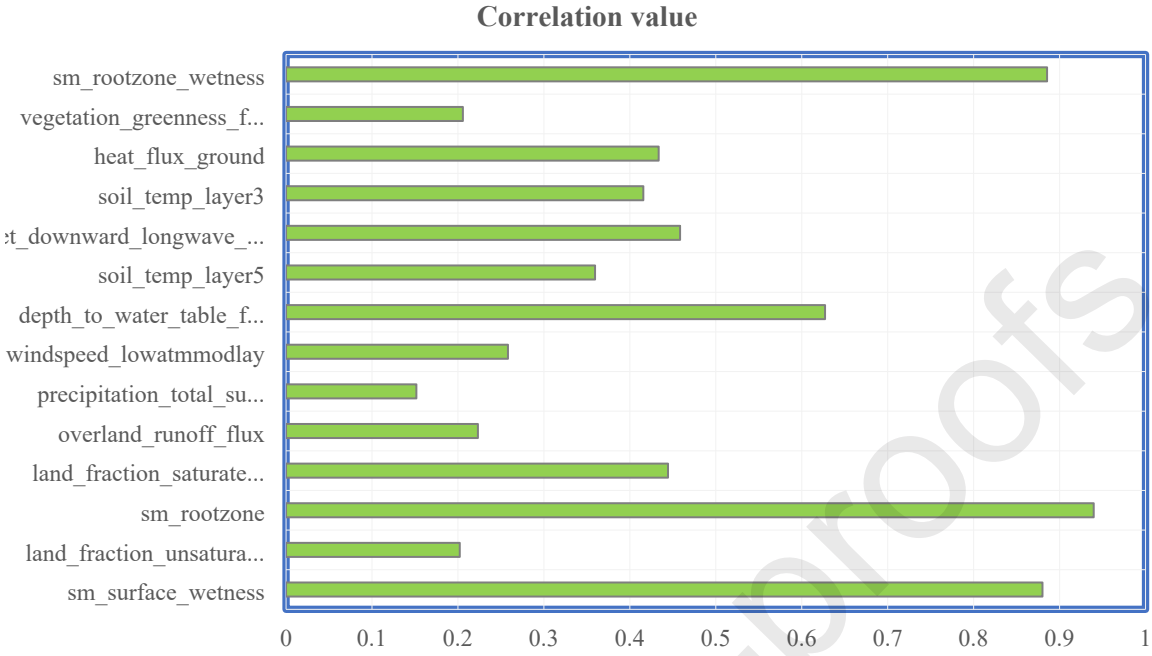


Fig. 4. Correlation value of highly important 14 features with sm_surface.

The GBR model was implemented using the scikit-learn library (Pedregosa et al., 2011). Hyperparameter tuning was systematically conducted using the GridSearchCV function. The parameters optimization includes, number of trees (n_estimators = 100), learning rate (learning_rate = 0.1), tree depth (max_depth = 3), and random state (random_state = 42). **Fig. 5** shows the general architecture of the GBR model. Setting a shallow tree depth (max_depth = 3) helped prevent overfitting, along with the benefit of feature selection in reducing model complexity. The training and evaluation process followed an 80:20 split, where 80% of the available data samples were used for training and 20% were used for testing. A single GBR model was trained and tested on data of each seven locations (Bhuj, Idar, Ankleshwar, Bhavnagar, Jamnagar, Vapi, and Palanpur) independently then additionally saved model on each location is tested on different location data to cross validate model's predictive ability. This cross-validation step helped to confirm that the model works well even when applied to different environmental conditions. Three common evaluation metrics, MSE, mean absolute error (MAE), and R-squared (R^2) were used to assess how well the model performed.

3.5 Details of Implementation

Linear Regression, Lasso, RF and GB were implemented in *Google Colab* with Python 3.8 and scikit-learn version 0.24.2. The environment provided a consistent and accessible platform for implementing and testing all models. GBR was tuned using GridSearchCV with 5-fold cross-validation. The final model used n_estimators = 100, learning_rate = 0.1, max_depth = 3, and random_state = 42. RF Regressor was tuned using n_estimators = 100 and max_depth = None

(i.e., fully grown trees). For Lasso Regression regularization parameter alpha was determined via built-in cross-validation to ensure optimal shrinkage without manual tuning. And Linear Regression was implemented using default scikit-learn settings.dom_state = 42.

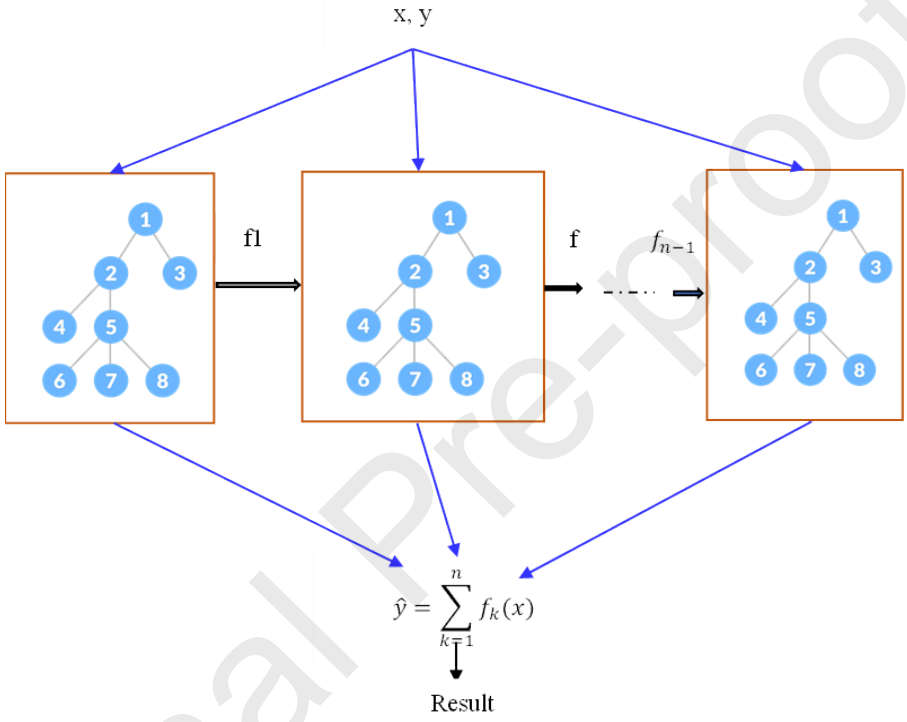


Figure 5. Gradient Boosting Regressor model.

4. Soil Moisture Prediction Process

The model generation process for assessing soil moisture levels using the SMAP geophysical dataset is explained as follows. The powerful GBR, which employs ensemble learning to measure typical interdependency present in the data, is the foundation of the suggested prediction model. Using weaker learning algorithms that have been separately taught from several models, the ensemble machine learning algorithm combines the findings in some way, usually by averaging all the predictions to produce an overall better prediction. This has proven to be effective, and a lot of study in the field of ensemble learning has been done on it (Hegazi et al., 2021). It is highly helpful to predict soil moisture when a subset of geophysical properties is chosen. When the model's hyperparameters are set appropriately, the model's performance can potentially increase dramatically. One technique for choosing a model's ideal hyperparameter is GridSearchCV (Ahmad et al., 2022). A comprehensive approach is employed to maximize the GBR

hyperparameters, instead of choosing the estimators or random state values at random, GridSearchCV's power is used to systematically identify the best suited values. After a thorough investigation, it was concluded that 100 estimators and a random state value of 42 would make up the optimal design. GridSearchCV enables us to explore a wide variety of hyperparameter combinations in a time efficient manner, which helps to ensure that proposed model is appropriately calibrated for the activity at hand. The utilized GridSearchCV methodology for model training and hyperparameter selection demonstrated in **Fig. 6**. The performance metrics have been evaluated during the whole model-training process which are R^2, MSE and MAE. These measures offered helpful information on how well the model was able to absorb the data fed to it during training. It is important to highlight that GBRs are resistant to overfitting by nature. However, a very good balance between bias and variance is achieved in the suggested model by carefully choosing the hyperparameters, as suggested by GridSearchCV. The suggested model results, after these modifications, will provide accurate and dependable forecasts for the SMAP soil moisture dataset. The trained model's strong prediction abilities are demonstrated by evaluation metrics. A prediction accuracy metric supplied by MSE, where higher performance is shown by lower numbers. R^2 metric also quantifies the percentage of variance in the objective variable (soil moisture) that the model accounts for. Performance on the training dataset provides a good starting point for the model's potential in the actual world.

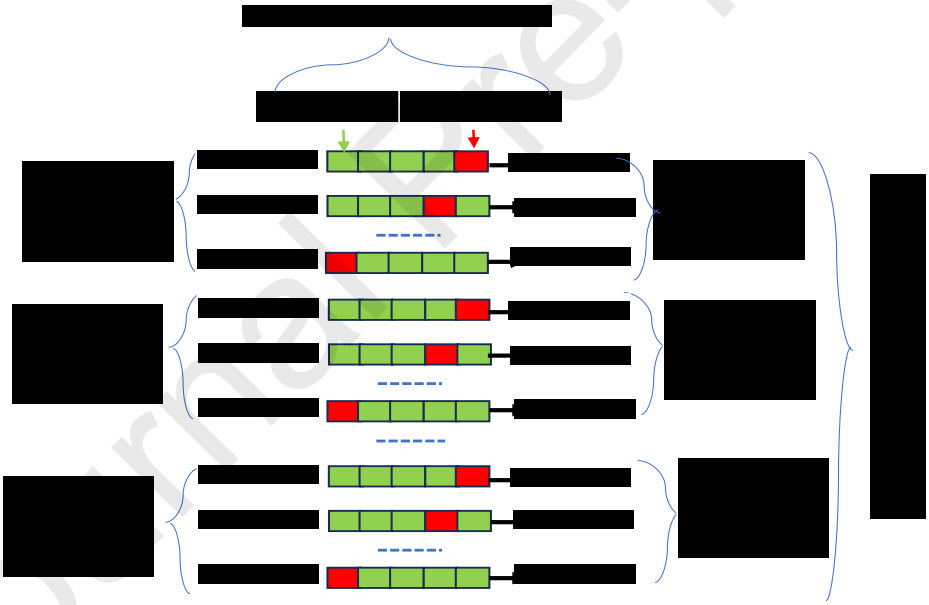


Fig. 6. Tuning of hyperparameters using fivefold cross-validation (GridSearchCV).

Basically, the MSE, R^2 and MAE are calculated using following expression (Chicco et al., 2021)

$$MSE = \frac{1}{n}\sum(\hat{p}_i - \overline{p})^2 \quad (5)$$

where, $\overline{p}$ = actual value of the target and $\hat{p}$ = predicted value of the target.

$$R^2 = \frac{SSR}{SST} = \frac{\sum(\hat{p}_i - \overline{p})^2}{\sum(p_i - \overline{p})^2} \quad (6)$$

where SSR represents the sum of squares regression and SST represents the sum of squares total.

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|x_i - x| \quad (7)$$

where, x_i = predicted value of target and x = actual value of target. A completely trained version of the model has been developed and saved as a file using the Python Pickle module, ensuring the model's accessibility and practicality in real-world applications. It guarantees that the suggested model's predictive ability can be applied to a range of activities concerning the estimation of moisture content in soil. The workflow of the proposed model is presented in **Fig. 7**. A thorough analysis has been done on the suggested model's outcome to determine the moisture content in soil, which was constructed using the SMAP geophysical dataset. Two key aspects are under consideration: the relationship between features and their significance, as well as the efficacy of the suggested model. In **Fig. 8**, the significance of characteristics is displayed. Among the features that were selected, 'sm_rootzone,' 'sm_rootzone_wetness,' and 'zm_surface_wetness' stand have high ratings. These scores show that these characteristics significantly affect predictions of soil moisture. This finding emphasizes the significant contributions that these variables make to the dynamics of soil moisture. Making a feature association heatmap (**Fig. 9**) encourages study into the relationships between the several variables that were selected. There is little intercorrelation among the input features, which is a positive sign for the suggested model's robustness. This assessment covers seven different areas in Gujarat, each with its own climate. It enables us to offer insightful information about the suggested model's capacity for prediction across a range of geographical areas. In addition, the investigation covered important dates including January, March, May, and August. Performance measures for every place and time point were carefully evaluated, including MSE, MAE, and R^2. **Table 3** displays, for each month throughout the four-year period, the average value of MSE, MAE, and R^2 to give a more detailed view of the suggested model's performance. The remarkable consistency of these results across sites is indicative of the

resilience of the suggested model. **Fig. 10** displays the overall model performance for four different months of each year at seven different locations in Gujrat.

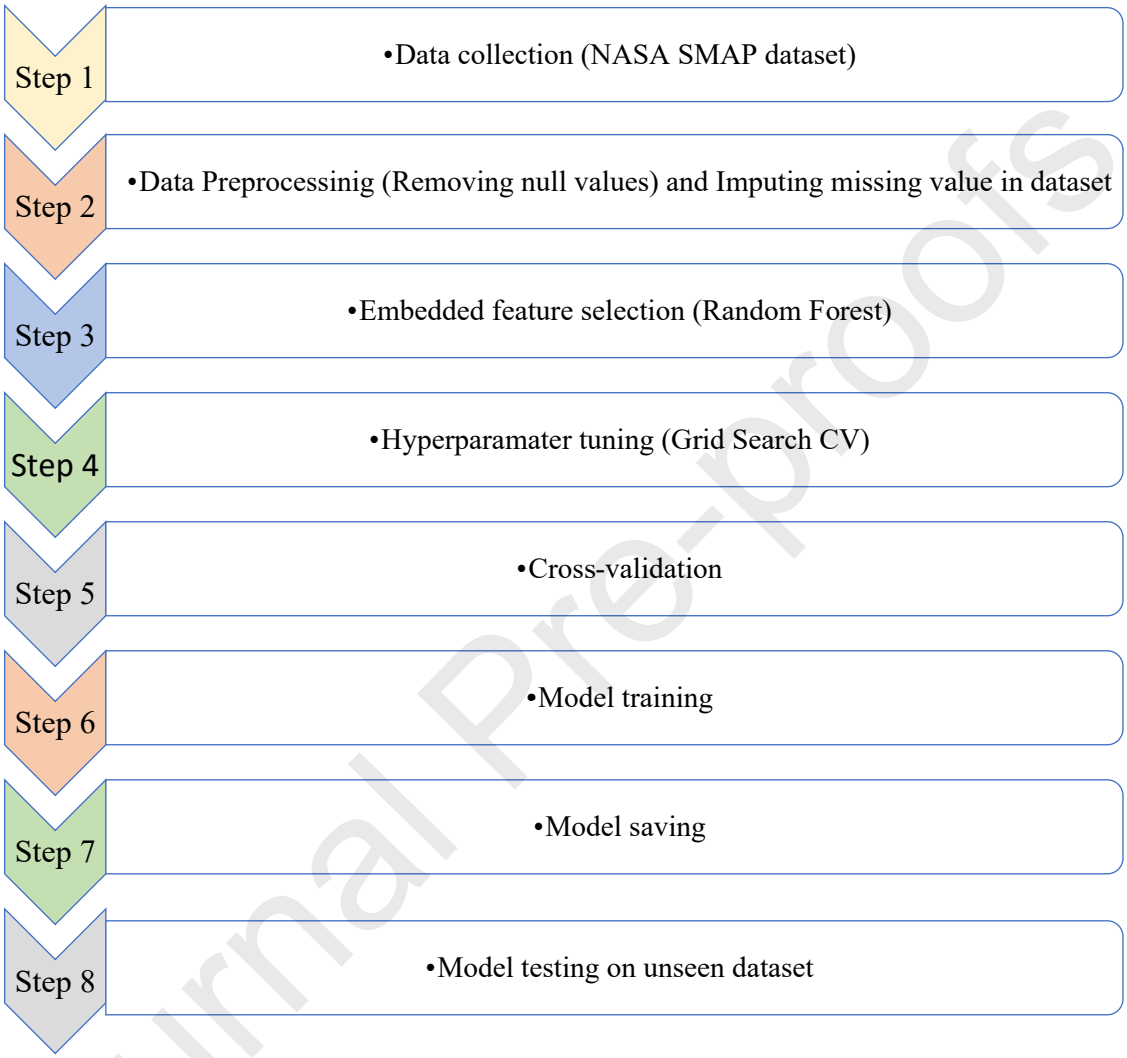


Fig. 7. Workflow of proposed model for soil moisture estimation.

Table 3 The average value of MSE, MAE, and R^2.

Months	**MSE** (m^3/m^3)	**MAE** (m^3/m^3)	**R^2**
January	4.8880×10^{-05}	0.0048	0.9972

March	4.7788×10^{-05}	0.0047	0.9974
May	5.2202×10^{-05}	0.0049	0.9971
August	5.3728×10^{-05}	0.0050	0.9971

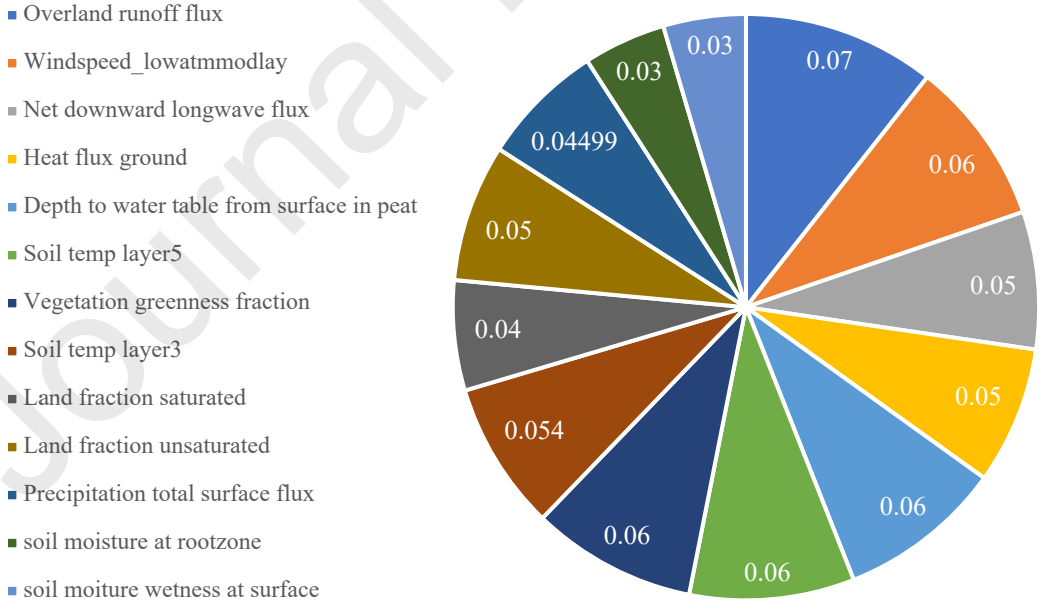


Fig 8. Importance score of features.

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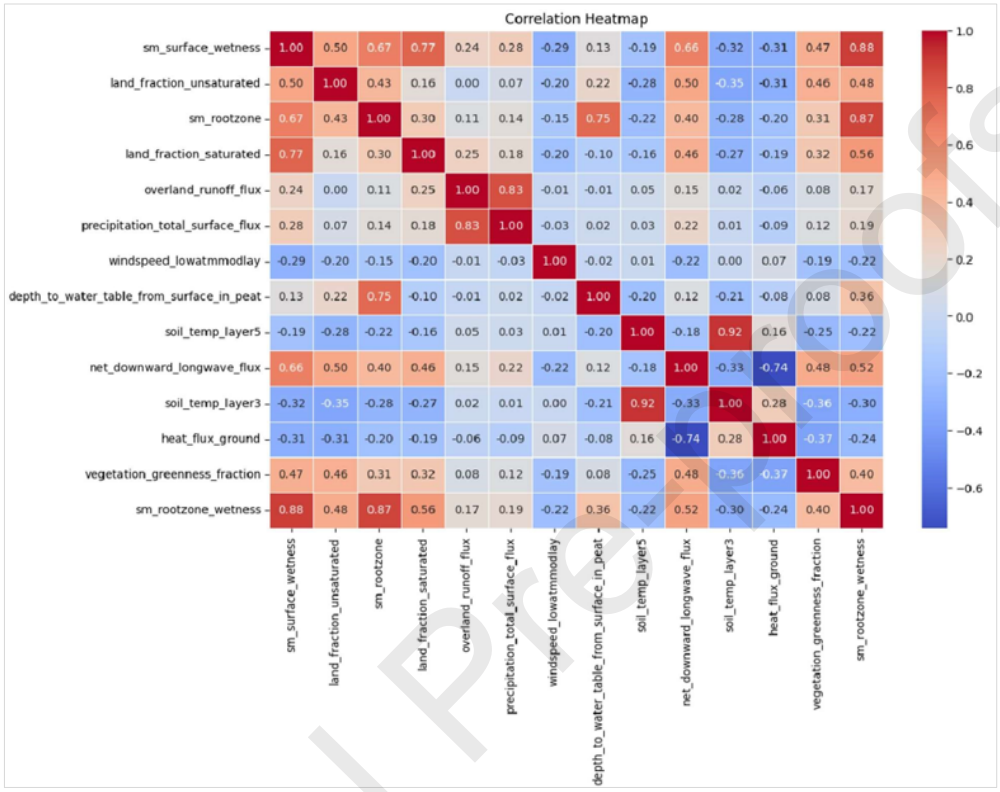


Fig 9. Correlation heatmap of selected feature.

5. Performance Analysis for Soil Moisture Prediction

There are several additional models which are commonly known as Linear Regression, RF, Regressor and Lasso Regression etc. These models are compared for analyzing the GBR performance. It's evident that the GBR model gave the greatest results. In **Fig. 11**, all methodologies' relative performance analysis is graphically displayed. While RF showed slightly lower MSE and MAE in some cases but both RF and GBR had comparable R^2 scores. However, GBR has shown a significantly better balance between accuracy and resource usage. During execution in Google Colab, RF has consumed about more than 10 GB of RAM just after fewer iterations as number of trees increases and sometimes it leads to crashing of kernel due resource exhaust while GB maintains accuracy as MSE of 4.8×10^{-5} m^3/m^3, MAE of 0.005 m^3/m^3, and R^2 of 0.99 m^3/m^3 with consummation of RAM below 8 GB always which proved it better in terms of accuracy and computational efficiency.

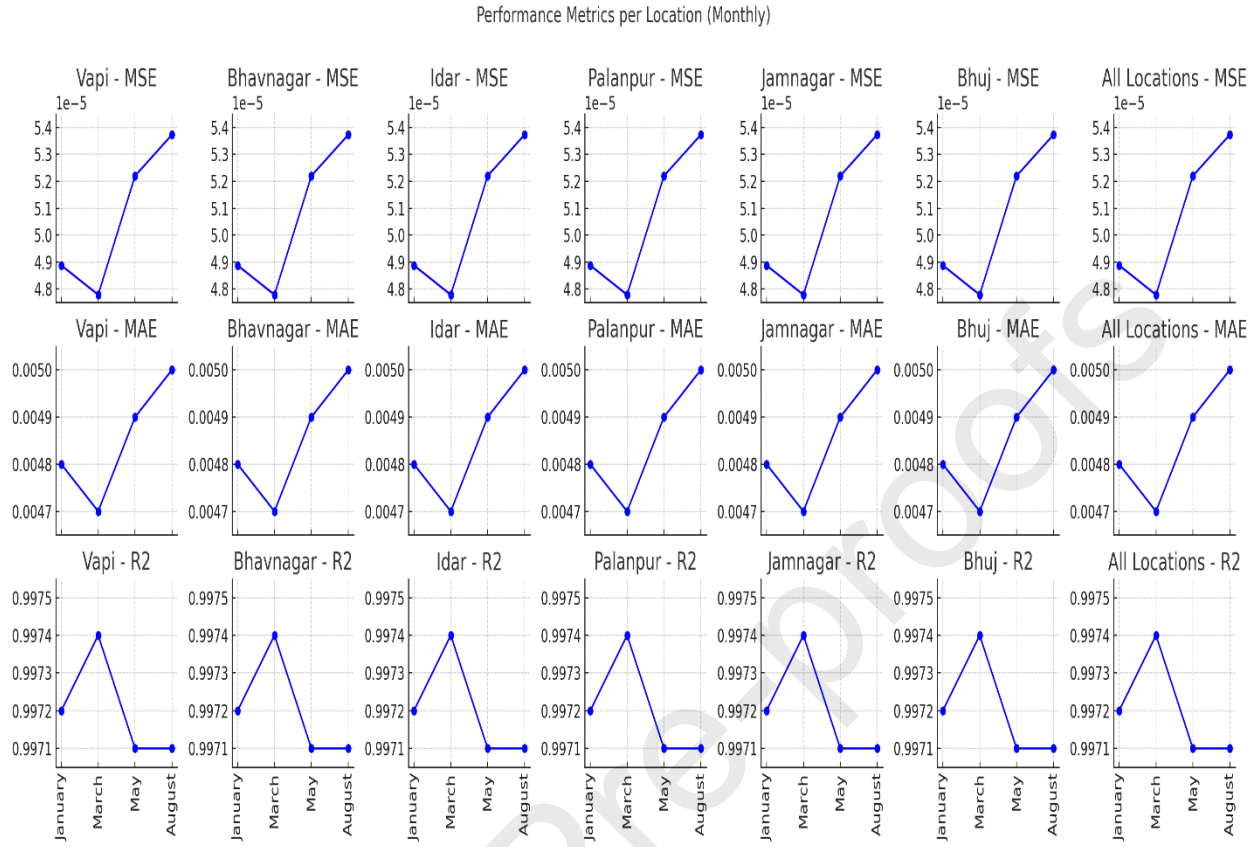


Fig 10. Performance of the proposed model in seven different locations.

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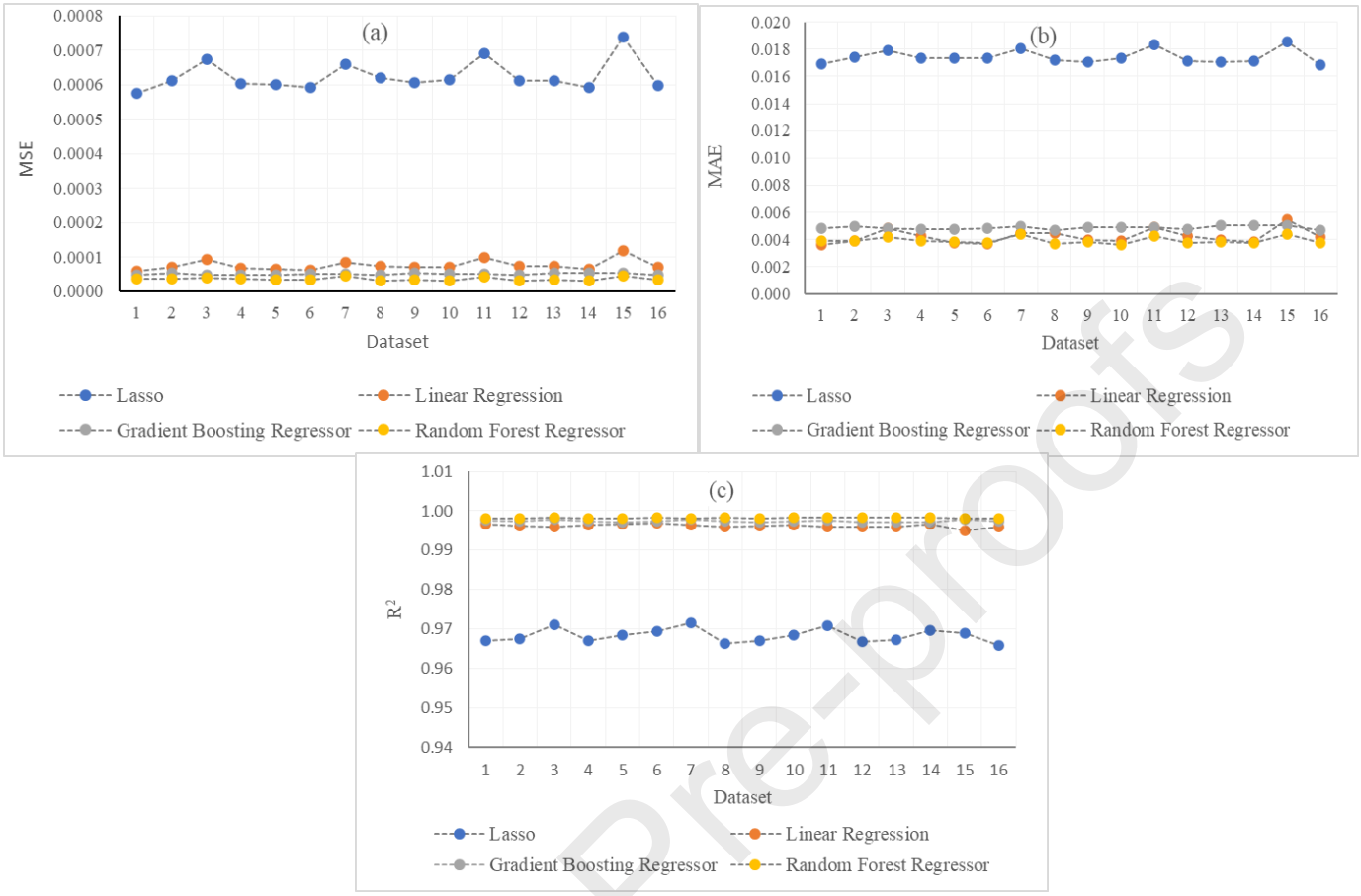


Fig. 11 (a)-(c) Comparative performance analysis of GBR and other regression methods.

Nevertheless, the GBR approach is flexible and dependable. By carefully adjusting the hyperparameters and carrying out real-world tests, the model's adaptability is shown, enabling it to meet the challenges of soil moisture prediction. These performance criteria confirm the GBR model's usefulness and establish it as an indispensable instrument for environmental monitoring, decision-making, and future study.

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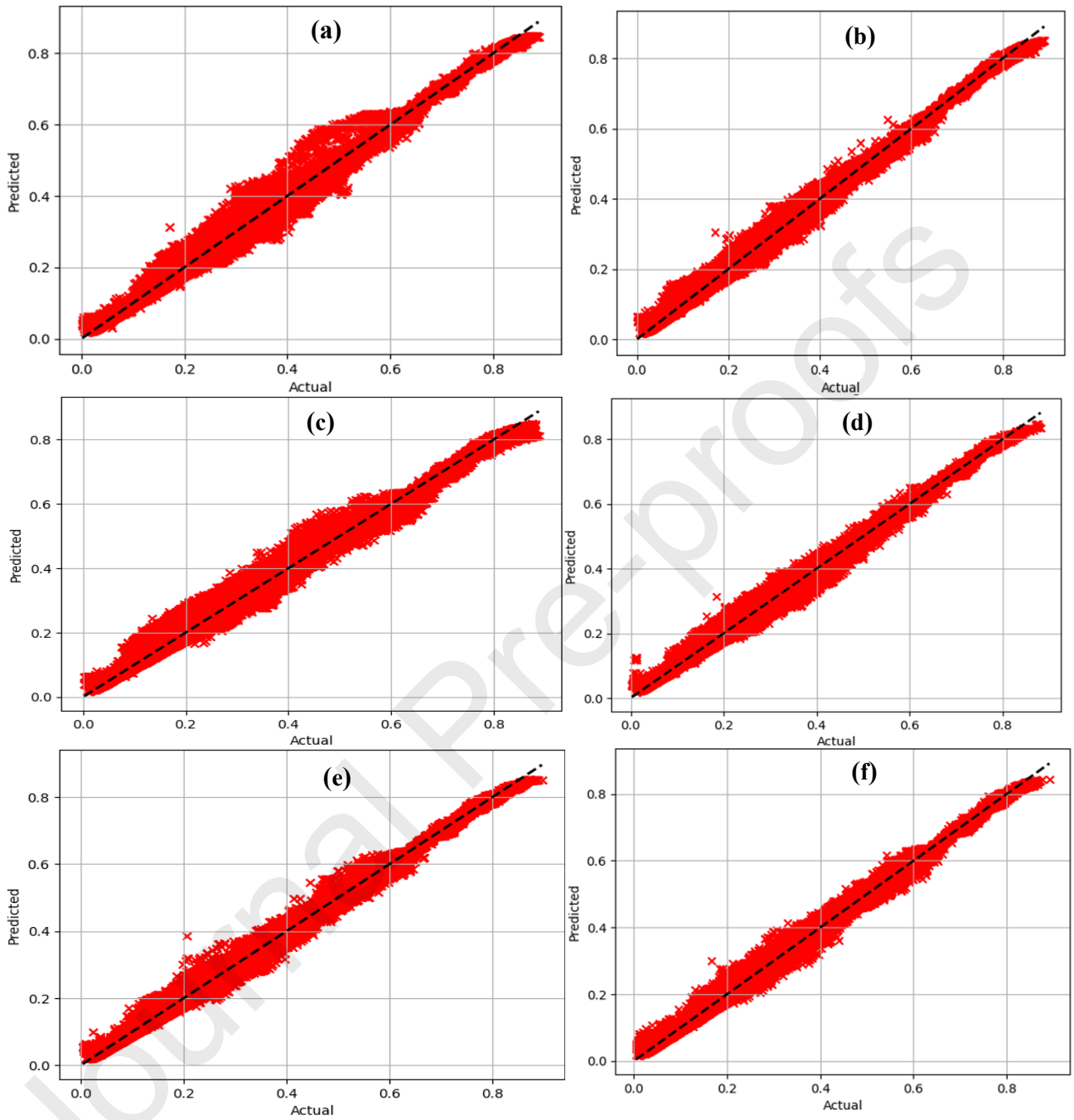


Fig. 12 Soil moisture prediction performance of GBR for (a) Bhuj, (b) Idar, (c) Ankleshwar, (d) Bhavnagar, (e) Jamnagar, and (f) Vapi.

The trained GBR model has been tested on all selected seven locations for the new dataset and the accuracy plot for actual verses predicted value of the soil moisture is being plotted (**Figs. 12 (a)-(f)**). It has been noticeable for all the locations that model has performed exceptionally well for all the dataset by predicting the values very near to the actual value of the moisture content of

soil. The level to which the model's predictions and actual values diverge is shown by points that deviate from the diagonal line. Points below the line indicate underestimate, while points above the line indicate overestimation by the model. When values fall on the diagonal line, it means that the moisture content value of soil is same for both actual and predicted values. The degree to which the model's predictions and actual values diverge is shown by points in **Figs. 12 (a)-(f)** that deviate from the diagonal line. The maximum divergence in actual values and predicted values is noticed for soil moisture between 0.2-0.6 for all locations. While the accuracy of prediction is higher for lower < 0.2 and higher > 0.2 soil moisture. On the basis of visual analysis, it is obvious that the maximum divergence is noticed in the prediction of moisture content of soil for Bhuj (**Fig. 12 (a)**) and the minimum divergence is achieved for Idar (**Fig. 12 (b)**) and Vapi (**Fig. 12 (f)**). There are a few observations in each of the datasets for which actual and predicted values of soil moisture deviate significantly.

If the prediction performance of the method for the locations Bhuj (**Fig. 12 (a)**, Ankleshwar (**Fig. 12 (c)**) and Jamnagar (**Fig. 12 (e)**) is observed closely, it has been seen that there is maximum divergence in between 0.2-0.6 than the other four locations (**Fig. 12 (b), Fig. 12 (d), and Fig. 12 (f)**). It is due to the unpredictable weather pattern at these locations which the model fails to accurately predict the moisture content of soil value near to the actual value. The same which the model has shown has also been mentioned in the official website of Indian Meteorological Department (Bentéjac et al., 2021) for these locations that these locations have the unpredictable weather pattern.

6. Discussion

This study proposes a novel surface soil moisture prediction model by utilizing GB Regressor which is trained on NASA's SMAP dataset (SMAP, 2015). Unlike many previous approaches over last 3 to 5 years that rely on satellite imagery and deep learning models like ConvLSTM or CNNs (Liu et al., 2022; Xu et al., 2021; Zhang et al., 2021), our approach focuses on structured tabular data. This model selection increases computational efficiency and model interpretability, which is very necessary in environmental applications where transparent decision-making is needed. The model achieved high accuracy across all seven selected regions in Gujarat, India. As shown in Table 3, the R^2 values consistently remained close to 0.99 m^3/m^3, with MSE values on the order of 10^{-5} m^3/m^3. This is a clear indication of strong learning and generalization by the model. Key features such as sm_rootzone, sm_rootzone_wetness, and sm_surface_wetness stood out as dominant predictors. Their importance was established through RF-based feature selection (**Fig. 8**) and validated by the correlation analysis (**Fig. 4**) (Speiser et al., 2019). To validate model performance spatially, predictions were tested across diverse locations—coastal (Bhavnagar, Jamnagar), industrial (Ankleshwar, Vapi), semi-arid (Bhuj), and arid (Palanpur). The soil moisture prediction plots for each location are shown in **Figs. 12 (a)-(f)**. The model's predictive curves align well with actual soil moisture values in most regions. However, for locations like Bhuj (**Fig. 12 (a)**), Ankleshwar (**Fig. 12 (c)**), and Jamnagar (**Fig. 12 (e**)), moderate divergence was observed in the mid-range moisture band (0.2–0.6). This can be explained by the unpredictable or dynamic weather conditions in these regions, as outlined in Table 1, including industrial interference or extreme semi-arid climates. These unpredictable signs show a need for more adaptive or hybrid modeling in such zones (Bentéjac et al., 2021). A comparative performance evaluation with other

regression models was conducted and visualized in **Fig. 11**. While RF occasionally yielded slightly better MSE or MAE, its resource usage was notably higher, often crossing 10 GB in RAM during training resulting in kernel crashes on Google Colab. Whereas GBR consistently maintained strong accuracy ($R^2 \approx 0.99$ m^3/m^3) with memory consumption of less than 8 GB. Thus, in practical applications, especially on constrained hardware, GBR provides a more reliable trade-off between performance and computational cost (Shrivastav & Jha, 2021; Reichle, 2014; Pedregosa et al., 2011). Other than that, GBR model emphasizes interpretability and reproducibility, a gap often found in deep learning approaches (Suganeshwari & Divya, 2024). Using an embedded RF feature selection strategy (**Fig. 2**), users can trace which features influence the prediction. This transparency is crucial for scientific validation and policy implementation (Speiser et al., 2019). Additionally, our training procedure an 80: 20 train-test split and 5-fold cross-validation, as visualized in **Fig. 6** ensures reproducibility and robust hyperparameter tuning (Ahmad et al., 2022; Emmanuel et al., 2021). The ability of models to perform consistently in the varying environment for varying time shows efficiency in accuracy, low resource usage and transparent modelling. But there are few conditions like excess rain, flood etc. which affect model performance much by reducing its predictive power. Despite strong results, the model has some limitations. In complex environmental zones (e.g., coastal, or industrial), standard geophysical features may not fully capture short-term fluctuations. Like too much rainfall or industrial discharge are difficult to model using only static features. Therefore, future work can focus on (i) integrating temporal satellite data or real-time meteorological features (Prasad et al., 2018), adopting transfer learning for regional adaptation (Li et al., 2021), or developing meta-ensemble models for dynamic regions (Rao et al., 2019).

7. Conclusion

This study develops a highly accurate and reliable model using GBR to estimate soil moisture dynamics across seven diverse locations: Bhuj, Jamnagar, Bhavnagar, Vapi, Ankleshwar, Palanpur, and Idar. With an emphasis on four crucial testing months (January, March, May, and August), this study attempted to give a thorough understanding of soil moisture fluctuations across Gujarat's diverse geographic and climatic settings between 2020 and 2023. The GBR's remarkably low values of MSE and high values of R^2 demonstrate its ability to capture intricate soil moisture fluctuations across these diverse regions and timeframes. The incorporation of the MAE value for GBR is particularly significant because it further demonstrates the model's ability to make accurate predictions while minimizing prediction errors. In the accurate prediction of soil moisture,'sm_rootzone,"sm_rootzone_wetness,' and 'sm_surface_wetness' have emerged as crucial determinants. They open the door to more complex and successful methods of predicting soil moisture in further studies. The accuracy and stability of GBR have the potential to completely change how water resources are managed, how agriculture is conducted, and how climate change adaptation plans are implemented.

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Appendix A (Table A1): List of All 44 Geophysical Features Extracted from SMAP L4_SM Dataset.

Feature Name	Description
baseflow_flux	Subsurface water flow to rivers (GEOS-5 Modeled)
depth_to_water_table_from_surface_in_peat	Depth to groundwater table in peat (GEOS-5 Modeled)
free_surface_water_on_peat_flux	Surface water flux on peatlands (GEOS-5 Modeled)
heat_flux_ground	Ground-to-atmosphere heat exchange (GEOS-5 Modeled)
heat_flux_latent	Latent heat flux (GEOS-5 Modeled)
heat_flux_sensible	Sensible heat flux (GEOS-5 Modeled)
height_lowatmmodlay	Height of lower atmospheric model layer (GEOS-5 Modeled)
land_evapotranspiration_flux	Evapotranspiration rate (GEOS-5 Modeled)
land_fraction_saturated	Saturated land fraction (GEOS-5 Modeled)
land_fraction_snow_covered	Land snow cover fraction (GEOS-5 Modeled)
land_fraction_unsaturated	Unsaturated land fraction (GEOS-5 Modeled)
land_fraction_wilting	Land fraction under wilting (GEOS-5 Modeled)

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leaf_area_index	Leaf area per ground area (GEOS-5 Modeled)
mwrtm_vegopacity	Microwave vegetation opacity (Observed via radiometer)
net_downward_longwave_flux	Net incoming longwave radiation (GEOS-5 Modeled)
net_downward_shortwave_flux	Net incoming shortwave radiation (GEOS-5 Modeled)
overland_runoff_flux	Surface runoff rate (GEOS-5 Modeled)
precipitation_total_surface_flux	Total precipitation (GEOS-5 Modeled)
radiation_longwave_absorbed_flux	Absorbed longwave radiation (GEOS-5 Modeled)
radiation_shortwave_downward_flux	Incoming solar radiation (GEOS-5 Modeled)
sm_profile	Soil moisture profile (GEOS-5 Modeled)
sm_profile_pctl	Soil moisture profile percentile (GEOS-5 Modeled)
sm_profile_wetness	Soil moisture profile wetness index (GEOS-5 Modeled)
sm_rootzone	Rootzone soil moisture (0–100 cm) (GEOS-5 Modeled)
sm_rootzone_pctl	Rootzone moisture percentile (GEOS-5 Modeled)

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sm_rootzone_wetness	Rootzone wetness index (GEOS-5 Modeled)
sm_surface	Surface soil moisture (0–5 cm) (Observed by SMAP radiometer)
sm_surface_wetness	Surface wetness index (GEOS-5 Modeled)
snow_depth	Snowpack depth (GEOS-5 Modeled)
snow_mass	Snow mass (GEOS-5 Modeled)
snow_melt_flux	Snowmelt runoff flux (GEOS-5 Modeled)
snowfall_surface_flux	Snowfall flux (GEOS-5 Modeled)
soil_temp_layer1	Soil temp 0–5 cm depth (GEOS-5 Modeled)
soil_temp_layer2	Soil temp 5–15 cm depth (GEOS-5 Modeled)
soil_temp_layer3	Soil temp 15–35 cm depth (GEOS-5 Modeled)
soil_temp_layer4	Soil temp 35–60 cm depth (GEOS-5 Modeled)
soil_temp_layer5	Soil temp 60–100 cm depth (GEOS-5 Modeled)
soil_temp_layer6	Soil temp 100–200 cm depth (GEOS-5 Modeled)
soil_water_infiltration_flux	Soil water infiltration rate (GEOS-5 Modeled)
specific_humidity_lowatmmodlay	Specific humidity at low atmosphere (GEOS-5 Modeled)

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surface_pressure	Surface atmospheric pressure (GEOS-5 Modeled)
surface_temp	Surface land temperature (GEOS-5 Modeled)
temp_lowatmmodlay	Near-surface air temperature (GEOS-5 Modeled)
vegetation_greenness_fraction	Green vegetation cover fraction (GEOS-5 Modeled)
windspeed_lowatmmodlay	Near-surface wind speed (GEOS-5 Modeled)

Appendix A (Table A2): Description of SMAP-L4 Dataset Fields.

Dataset Name	**Description**
Geophysical Data	Contains environmental variables like soil moisture, land temperature, vegetation, surface fluxes, etc., used for prediction.
EASE2 Global Projection	Using Equal-Area Scalable Earth Grid, it gives grid and projection information
Cell_Latitude	Latitude values of each grid cell.
Cell_Longitude	Longitude values of each grid cell.
Cell_Column / Cell_Row	Grid-based positional indices of each cell.
Time	Timestamps for each data capture instance at 3-hr interval.
X / Y Coordinates	Additional spatial coordinate metadata used in grid projection context.

Metadata	It contains information about data present.

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Declaration of interests

☒ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

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